

Shrey Patel

shrey.anup.patel@gmail.com | shreypatel.uk | github.com/Shrey-Patel-05 | +44-7483367887

Education

University of Manchester — Top 12% in Cohort of 370

Bachelor's in Computer Science — Year 1: 80.3%; Year 2: 76.3% (First-Class)

Modules: Software Engineering, Data Structures & Algorithms, Data Science

Manchester, UK

Sep 2024 – Jun 2027

Langley Grammar School

A Level: Mathematics, Computer Science, Further Mathematics, History; A*A*AA

GCSE: 10 Grade 9s, 1 Grade 8

Slough, UK

Sep 2017 – Jun 2024

Experience

Tripadvisor

Software Engineering Intern — Payments

London, UK

Jun 2026 – Sep 2026

- **System Design & Ownership:** Designed and delivered a Spring Boot service processing hundreds of supplier-verification events daily, replacing a legacy Lambda workflow with a maintainable, event-driven architecture.
- **Payouts Modernisation:** Defined transaction flows for a new payouts platform handling \$2.5B annually, defining APIs and extending an existing credits system to support transfers between accounts.

GreatUniHack & Student Hack

Developer

Manchester, UK

Mar 2025 – Oct 2025

- **Frontend Development:** Built modular React/Next.js components for hackathon websites, including custom particle effects, Framer Motion animations, and Three.js planetary models.
- **Event & Community Support:** Volunteered onsite at 150+ attendee hackathons and moderated Discord channels, facilitating communication between participants, sponsors, and organisers.

Code Ninjas

Code Sensei

Slough, UK

Feb 2023 – Jun 2024

- **Mentorship:** Oversaw groups of 10–15 children, ages 5–16, during coding workshops focused on elementary programming skills in JavaScript and C#.
- **Assessment & Student Progress:** Graded 50+ student projects, asynchronously and in-person, providing detailed feedback on game quality and debugging skills.

Projects

Lego Identification & Substitution Software

- Engineered a computer vision pipeline, enabling LEGO brick scanning with OpenCV; trained CNN on 40,000 images to classify 50 brick types with 75%+ accuracy.
- Developed a hash table-based inventory system, underpinned by a bespoke greedy LEGO part optimisation algorithm for efficient set building and brick substitutions.

Movie Recommendation Website

- Developed a responsive React frontend with Material-UI, implementing dynamic questionnaires and Tinder-style swiping mechanics for prospective films in batches of 20.
- Collaborated with a team of four to integrate asynchronous API communication with a Flask backend, handling user authentication, session management, and data persistence.

Simplex Method Software

- Created a linear programming optimisation engine, featuring interactive tableau visualisations and dynamic constraint management, implementing both standard and two-phase simplex algorithms.

Technical Skills

- **Languages:** Java, Python, TypeScript, SQL, C#, HTML/CSS
- **Frameworks & Technologies:** Spring Boot, React, Next.js, TensorFlow/Keras, OpenCV, AWS (ECS, ECR, DynamoDB, SNS/SQS), Docker, Grafana, GitLab CI/CD

Societies & Activities

- **Societies:** Softball — regular training, twice weekly; Robotics — built an autonomous maze-traversing buggy.
- **Hobbies:** Guitar, Baseball, Classic Cinema